

Hospital Information & Financial Management System

Comprehensive Accounting Glossary, Double-Entry Rules, and Supply Chain to FMS Data Integration Architecture

Part 1: Supply Chain & Inventory to FMS Data Connection Architecture

END-TO-END
WORKFLOW

SUPPLY CHAIN & INVENTORY MODULES

- Inventory (IMS):** Stock level alert triggered at reorder threshold
- Procurement (PSM):** Purchase Requisition (PR) generated
- PO Management:** PO #2026-104 dispatched with Net 30 terms
- Warehousing (SWS & DTRS):** Goods received, counted, GRN issued

FINANCIAL MANAGEMENT SYSTEM (FMS)

- Budget Gatekeeper:** Verify & freeze budget encumbrance (₱100k)
- Accounts Payable (AP):** Ingest expected liability & terms
- 3-Way Match Verification:** Verify PO ↔ GRN ↔ Vendor Bill
- GL & Treasury:** Post inventory liability & release EFT payment

REAL-WORLD CLINICAL CASE SCENARIO

Pharmacy Reorder: 1,000 vials of IV Antibiotics from *Unilab* valued at **₱100,000.00** (₱100/vial, Net 30).

STEP 1: STOCK ALERT & PR

IMS → PSM

Automated Requisition

Pharmacy antibiotic stock drops below safe threshold (<50 vials). IMS triggers an automated Purchase Requisition (PR) to hospital procurement officers.

Payload: Item #ABX-1G, Qty: 1000, Dept: Pharmacy

STEP 2: BUDGET GATEKEEPER

PSM → FMS Budget

Encumbrance Reservation

FMS validates available fiscal budget: "Does Pharmacy have ₱100,000 remaining in 2026 Medicine Budget?" Freezes funds into Budget Encumbrance so it cannot be double-spent.

Status: ₱100,000 Encumbered / Reserved

STEP 3: PURCHASE ORDER PUSHED

Procurement → AP

Official PO Generation

Procurement issues PO #2026-104 to Unilab with Net 30 terms. Digital copy automatically synchronizes to FMS Accounts Payable for projected liability scheduling.

Terms: Net 30 Days | Total: ₱100,000.00

STEP 4: DELIVERY & WAREHOUSE

SWS → DTRS → AP

Goods Receipt (GRN)

Delivery arrives at loading dock. SWS counts 1,000 vials, inspects expiry/batch, increments live inventory, and generates Goods Receipt Note (GRN) sent to AP.

Stock: +1,000 vials | GRN #GRN-2026-882

STEP 5: 3-WAY MATCH

FMS AP Engine

Three-Way Verification

Compares PO (1,000 @ ₱100), GRN (1,000 @ ₱100), and Vendor Bill #UNI-9821 (₱100,000). Exact match automatically clears bill for posting and holds variances.

PO (₱100k) = GRN (₱100k) = Bill (₱100k) ✓

STEP 6: RECORDING DEBT

FMS General Ledger

Double-Entry Posting

Recognizes medical asset and hospital accounts payable obligation:

- Debit (Asset):** Medical Inventory ₱100,000
- Credit (Liability):** Accounts Payable ₱100,000

Balanced Journal Entry #JE-AP-2026-401

STEP 7: BANK PAYMENT RELEASE & LIQUIDATION

FMS Disbursement → Cash Management

Settlement via Electronic Fund Transfer (EFT) or Check

On Day 30, Finance generates Disbursement Voucher (DV #DV-2026-550). Approved EFT transfer sends ₱100,000 from hospital bank account to Unilab. Budget encumbrance is liquidated, and Accounts Payable is cleared:

Debit: Accounts Payable ₱100,000 (Debt extinguished) | **Credit:** Cash in Bank ₱100,000 (Asset reduced)

Supply Chain to FMS Cross-Module Integration Summary

STAGE	SUPPLY CHAIN SUB-MODULE	FMS SUB-MODULE	DATA EXCHANGED / LEDGER IMPACT
1. Budget Check	Procurement (PSM)	Budget Management	Estimated Cost → Budget Encumbrance Reservation
2. Order Placed	Purchase Order Management	Accounts Payable (AP)	PO Number, Vendor ID, Item Prices, Expected Delivery
3. Delivery	Smart Warehousing (SWS)	Accounts Payable (AP)	GRN / Receiving Report, Actual Quantity Delivered
4. Verification	Document Tracking (DTRS)	Accounts Payable (AP)	Vendor Invoice → 3-Way Match Verification
5. Financial Record	(Automated by FMS)	General Ledger (GL)	Debit Medical Inventory (Asset) vs. Credit AP (Liability)
6. Cash Payout	(Automated by FMS)	Disbursement Management	Debit AP vs. Credit Cash in Bank → Bill marked PAID

1. The Core 5 Financial Account Categories (Chart of Accounts Foundation)

Asset ACCOUNT TYPE: 1XXX Economic resources owned by the hospital expected to yield future economic benefits. Hospital Example: Cash in Bank, Medical Inventory (antibiotics, implants), Hospital Buildings, MRI/CT scanners, Patient Accounts Receivable.	Liability ACCOUNT TYPE: 2XXX Present financial debts or obligations arising from past transactions to be settled with hospital assets. Hospital Example: Accounts Payable to pharmaceutical vendors (Unilab), Unremitted Doctor Professional Fees, BIR Taxes Withheld (EWT/VAT).
Equity ACCOUNT TYPE: 3XXX The residual interest in hospital assets after deducting all liabilities (Assets – Liabilities = Equity). Hospital Example: Hospital Founder Capital Contributions, Retained Earnings accumulated from prior fiscal years, Capital Reserves.	Revenue (Income) ACCOUNT TYPE: 4XXX Gross inflow of economic benefits earned from healthcare services rendered and hospital operations. Hospital Example: Inpatient Room & Board, ER Triage Fees, Laboratory & Diagnostic Testing, Pharmacy Drug Sales, Operating Room Charges.
Expense ACCOUNT TYPE: 5XXX Operational costs and outflows consumed during the accounting period to deliver healthcare services and generate revenue. Hospital Example: Clinical Staff & Nurse Salaries, Facility Electricity & Oxygen Gas utility bills, Disposable Surgical Supplies, Ambulance Maintenance.	

2. The Mechanics of Bookkeeping & Double-Entry Engine

Debit (Dr) LEFT SIDE An accounting entry on the left side of a ledger. Increases Assets and Expenses; Decreases Liabilities, Equity, and Revenue. Example: When receiving ₱5,000 cash from a patient at the cashier desk, <i>Cash (Asset)</i> is Debited (+₱5,000).	Credit (Cr) RIGHT SIDE An accounting entry on the right side of a ledger. Increases Liabilities, Equity, and Revenue; Decreases Assets and Expenses. Example: When rendering a ₱5,000 lab blood panel, <i>Laboratory Revenue (Income)</i> is Credited (+₱5,000).
Debit-Credit Invariance Rule Σ DR === Σ CR The non-negotiable mathematical law of double-entry accounting: Total Debits must exactly equal Total Credits. Engine Rule: Zero-tolerance validation using PHP bcmath (scale 4). Any unbalanced transaction is blocked and rolled back.	General Ledger (GL) MASTER BOOK The master central repository containing all Chart of Accounts and the chronological financial journal entries of the hospital. System Feature: Real-time balance updates, debit-credit audit streams, and trial balance generation.
Journal Entry (JE) TRANSACTION UNIT A formal financial transaction record with a unique identifier, entry date, narration, and balanced debit/credit lines. System Feature: Auto-generated by billing, cashier drops, AP approvals, and disbursement releases.	Reversal Entry AUDIT IMMUTABILITY A compensating entry posted to nullify an erroneous posted entry without deleting historical records. Integrity Rule: Posted records are immutable. Corrections strictly generate an inverted JE marked REVERSED.

3. Patient Billing & Accounts Receivable (AR)

Accounts Receivable (AR) RECEIVABLE ASSET Uncollected monies owed to the hospital by discharged patients, PhilHealth, private HMO insurers, and government assistance funds. Hospital Scope: Tracked per patient encounter, HMO provider billing statement, and pending PhilHealth claims.	AR Aging (Aging Buckets) CREDIT ANALYSIS Classification of unpaid invoices based on days outstanding to assess credit risk and bad debt allowances. Standard Buckets: Current (0–30 days), 31–60 days, 61–90 days, 91–120 days, and Over 120 days (Delinquent/High Risk).
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Statement of Account (SOA)

OFFICIAL BILL

Comprehensive billing summary provided upon patient discharge displaying gross charges, statutory discounts, third-party covers, and balance.

Compliance: Formatted for hospital cashiering, patient clearance, and HMO reimbursement submission.

Copay / Out-of-Pocket

PATIENT SHARE

Net patient responsibility payable directly at the cashier desk after exhausting PhilHealth, HMO LOA, and Malasakit assistance.

Calculation: Gross Bill – (VAT Exemption + 20% Discount + PhilHealth ACR + HMO LOA + Guarantee Letters).

Third-Party Payers

INSURANCE / GOV

Entities settling medical charges on behalf of patients: PhilHealth ACR (All-Case-Rate), Private HMOs (Maxicare, Intellicare), and Government GLs.

Mechanism: Requires approved Letter of Authorization (LOA) or Malasakit Center Guarantee Letter (GL) from PCSO/DSWD.

Statutory Discounts (RA 9994 / 10754)

PH MANDATE

Philippine laws granting Senior Citizens (RA 9994) and PWDs (RA 10754) 12% VAT exemption plus 20% statutory discount on medical services & drugs.

Formula: Net of VAT = Gross / 1.1200; VAT Relief = Gross – Net; Discount = Net × 0.2000. Total = VAT Relief + Discount.

4. Supplier & Accounts Payable (AP)

Accounts Payable (AP) LIABILITY: 2010 Short-term liabilities the hospital owes to external suppliers for pharmaceutical items, surgical devices, and operational supplies. Control: Tracked by vendor aging, invoice due dates, and 3-way match validation.	Purchase Bill (Vendor Invoice) BILLING DOCUMENT The vendor's formal invoice demanding settlement for delivered goods or services, containing terms and tax breakdown. Hospital Example: Unilab Invoice #UNI-9821 requesting payment for 1,000 antibiotic vials under Net 30 terms.
3-Way Matching Engine FRAUD PREVENTION Automated verification comparing Purchase Order (PO) ↔ Goods Receipt Note (GRN) ↔ Vendor Purchase Bill. Validation: Ensures quantities and prices match exactly before an invoice can be approved for payment disbursement.	Payment Terms CREDIT TERMS Agreed contractual timeline for paying vendor obligations without penalties (e.g., Net 15, Net 30, Net 60 days). Cash Flow Impact: Drives AP aging schedules and automated treasury disbursement planning.

5. Disbursement & Cash Management

Disbursement FUND OUTFLOW The formal payout of hospital liquid funds to settle vendor invoices, payroll, doctor professional fees, or operational costs. Execution: Executed through check printing (CheckRegister) or online Electronic Fund Transfers (EFT).	Disbursement Voucher (DV) APPROVAL DOCUMENT Internal legal voucher signed by the Finance Director and Comptroller approving disbursement against validated purchase bills. Audit Requirement: Contains accounting distribution (debit/credit), supporting attachments, and BIR 2307 deductions.
Petty Cash Fund (Imprest System) FIXED CASH POOL A fixed, revolving cash fund (e.g., ₱10,000) for minor emergency expenses that is replenished periodically against approved receipts. Hospital Use: Emergency courier postage, emergency lab specimen transport, minor patient transport assistance.	Bank Reconciliation TREASURY CONTROL Periodic process of reconciling the hospital's general ledger cash accounts against official electronic bank statements. Reconciliation Items: Outstanding checks, deposits in transit, bank service charges, interest earned.

6. Budgeting & Fiscal Control

Fiscal Budget Allocation ANNUAL CAP Authorized departmental spending ceiling approved for a fiscal year (e.g., ₱5,000,000 for Pharmacy).	Expanded Withholding Tax (EWT) TAX DEDUCTION Mandatory tax deducted from payments to goods vendors (1%), service vendors (2%), and Doctor Professional Fees (10%/15%).
Budget Encumbrance RESERVATION Freezing budget funds upon Purchase Order issuance to prevent over-obligation prior to delivery.	BIR Form 2307 TAX CERTIFICATE Certificate of Creditable Tax Withheld at Source issued to doctors and suppliers for BIR tax credits.
Liquidation SETTLEMENT Unfreezing the encumbrance and recording the actual expenditure when the bill is settled.	CAS Audit Trails BIR MANDATE Tamper-proof, immutable activity logging (cas_audit_trails) required by the BIR Computerized Accounting System guidelines.

8. Financial Statements & Executive Reports

Trial Balance MATHEMATICAL VERIFICATION Internal worksheet listing ending debit and credit balances of all ledger accounts to guarantee ledger integrity before closing. Verification: If Total Debits ≠ Total Credits, closing is blocked and discrepancies flagged.	Balance Sheet (Financial Position) POINT-IN-TIME Statement displaying hospital Assets, Liabilities, and Equity on a specific date, illustrating overall financial solvency. Equation: Total Assets = Total Liabilities + Total Equity.
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Profit & Loss (P&L / Income Statement)

PERIOD SUMMARY

Summary of operational revenues and expenses over a specified period to show net profit or deficit.

Equation: Net Operating Income = Total Revenues - Total Operational Expenses.

Cash Flow Statement

CASH MOVEMENTS

Report tracking liquid inflows and outflows segregated into Operating, Investing, and Financing activities.

Liquidity: Proves whether operating collections are sufficient to support hospital capital expenditures and debt service.